1. Replace the NaN values with correct value and justify why you have chosen the same.

The salary column contains missing values for students who were not placed. These missing values can be replaced with 0 because students who were not placed did not receive any salary. Therefore, replacing NaN values with 0 is the most suitable approach

2. How many of them are not placed?

67 students

3. Find the reason for non placement from the dataset.

Students who were not placed has lower mark in SSC, HSC, Degree percentage, and entrance test when compared with placed students. Academic performance appears to be one of the major reasons for non-placement in the dataset given

4. What kind of relation between salary and mba_p?

salary and MBA percentage has weak positive relationship. As MBA percentage increases, salary tends to increase slightly but not very strong

5. Which specialization is getting minimum salary?

Mkt&HR specialization receives the minimum average salary when compared with Mkt&Fin

6. How many of them getting above 500000 salary?

3 students

7. Test the Analysis of Variance between etest_p and mba_p at significance level 5%.

Null Hypothesis (H0)- There is no significant difference between etest_p and mba_p. Alternative Hypothesis (H1) - There is a significant difference between etest_p and mba_p. Since the p-value is less than 0.05, the Null Hypothesis is rejected.

8. Test the similarity between degree_t (Sci&Tech) and specialisation (Mkt&HR) with respect to salary at significance level 5%.

Null Hypothesis (H0)- Salary distribution is similar between Sci&Tech students and Mkt&HR students. Alternative Hypothesis (H1)- Salary distribution differs between the two groups. Since the p-value is less than 0.05, the Null Hypothesis is rejected

9. Convert the normal distribution to standard normal distribution for salary column.

Use the Z-score formula

10. What is the Probability Density Function of the salary range from 700000 to 900000?

Use normal distribution formula whichgives the probability of a salary falling between ₹700000 and ₹900000

11. Test the similarity between degree_t (Sci&Tech) with respect to etest_p and mba_p at significance level of 5%.

Null Hypothesis (H0): -There is no significant difference between etest_p and mba_p for Sci&Tech students. Alternative Hypothesis (H1): There is a significant difference. Since the p-value is less than 0.05, the Null Hypothesis is rejected

12. Which parameter is highly correlated with salary?

The parameter etest_p (entrance testPercentage) shows the highest positive correlation with salary among the numerical variables.

13. Plot any useful graph and explain it.

The scatter plot graph shows a slight upward trend, indicating a weak positive relationship between MBA percentage and salary. Students with higher MBA scores tend to receive slightly higher salaries